Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 27: Assume the Opposite

Proof by Contradiction

Chartrand & Zhang, Section 3.7, pp. 108–111

Name: _____ Date: _____

Where we are — and a confession

This is the last proof technique in Chapter 3. It is also one you have **already seen twice**: Cantor's proof that $|A| < |\mathcal{P}(A)|$ and the diagonal argument that $[0, 1]$ is uncountable, both in Lecture 21, were proofs by contradiction. Today the method gets its name.

By the end of today you should be able to

1. Explain why deriving a contradiction proves a statement.
2. Write a proof by contradiction with a correct opening.
3. Follow and reproduce the proof that $\sqrt{2}$ is irrational.

1. Why it works

To prove a statement R: assume $\sim R$, and derive a contradiction C — something of the form $P \wedge (\sim P)$. Then we have established $(\sim R) \Rightarrow C$. Since C is false, the only way that implication can be true is if $\sim R$ is false — so R is true.

Lecture 7, doing the work

That step is **modus tollens**: from $(\sim R) \Rightarrow C$ and $\sim C$, conclude $\sim(\sim R)$, which is R by double negation.
And C is a contradiction in exactly the Week 3 sense — false in every row. Proof by contradiction is built entirely from tools you proved in the first three weeks.

The opening line

For $R : \forall x \in S,\ P(x) \Rightarrow Q(x)$, the negation is $\exists x \in S,\ P(x) \wedge (\sim Q(x))$. So begin:
"Assume, to the contrary, that there exists $x \in S$ such that $P(x)$ is true and $Q(x)$ is false."

2. A first example

Claim

There is no smallest positive real number.

Proof. Assume, to the contrary, that there is a smallest positive real number; call it r. Now consider $s = r/2$. Then s is a positive real number, and $s = r/2 \leq r$.

So s is a positive real smaller than r — contradicting that r was the smallest. ■

The move worth stealing

Once you assume an object exists, **give it a name**. Naming r is what let us build s from it. Proofs by contradiction often begin by naming the thing that supposedly exists and then using it against itself.

3. With parity

Result 3.48. Let $n \in \mathbb{Z}$. If $5n + 1$ is odd, then n is even.

Proof. Assume, to the contrary, that there is an integer n such that $5n + 1$ is odd and n is odd. Since n is odd, $n = 2k + 1$ for some integer k, so

$$5n + 1 = 5(2k + 1) + 1 = 10k + 6 = 2\big(5k + 3\big).$$

Since $5k + 3$ is an integer, $5n + 1$ is even — contradicting our assumption that it is odd. ■

Contradiction versus contrapositive

Compare this with a proof by *contrapositive* of the same result: assume n is odd, show $5n + 1$ is even. The algebra is identical.
The difference is in what you assume. **Contrapositive** assumes only $\sim Q$. **Contradiction** assumes P *and* $\sim Q$ together, and aims for any absurdity.
When the algebra is the same, the contrapositive is cleaner and usually preferred. Contradiction earns its keep when there is no implication to reverse — as in the next section.

4. The most famous proof in the subject

Theorem 3.49

The real number $\sqrt{2}$ is irrational.

There is no "if…then" here to take a contrapositive of. That is when contradiction is the natural tool.

We need one earlier result: **Theorem 3.24** — n^2 is even if and only if n is even. You proved it in Lecture 23.

Proof. Assume, to the contrary, that $\sqrt{2}$ is rational. Then $\sqrt{2} = a/b$ where $a, b \in \mathbb{Z}$, $b \neq 0$, and we may assume a/b is **reduced to lowest terms**.

Squaring, $2 = a^2/b^2$, so $a^2 = 2b^2$. Hence a^2 is even, and by Theorem 3.24, a is even. Write $a = 2c$ for some integer c.

Then $(2c)^2 = 2b^2$, so $4c^2 = 2b^2$, so $b^2 = 2c^2$. Hence b^2 is even, and by Theorem 3.24, b is even.

So a and b are both even, sharing a factor of 2 — contradicting that a/b was in lowest terms. ■

Where the contradiction came from

The contradiction is not with arithmetic. It is with an *assumption we chose to add* — that the fraction was reduced.
Chartrand calls this the subtle part: **it is hard to know in advance where the contradiction will appear**, and adding "in lowest terms" was a deliberate setup for it. That unpredictability is what makes proof by contradiction powerful and hard to plan.

Exit ticket

1. Write the opening sentence of a proof by contradiction of: "if n^2 is odd, then n is odd."

2. Why did the proof that $\sqrt{2}$ is irrational assume a/b was in lowest terms?

Before Monday

Read the Chapter 3 Highlights, pp. 112–113. They summarise every method in the chapter on one page.

Self-check (not collected). Section 3.7, Exercises 1, 3, 5, and 11 — odd-numbered, answers in the back. Exercise 11 is $\sqrt{3}$, the same argument as today.

Coming up

Homework 9 is due tonight, Friday, October 23, 11:59 p.m.
Quiz 8 is Monday, October 26 — Sections 3.4, 3.5, and 3.6.
Homework 10 covers Sections 3.4–3.7 and is due Friday, October 30.

Questions I still have